

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459076.6982 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.231 +/- 0.068
Transit depth (Rp/Rs)^2	5.34 +/- 3.12 [%]
Orbital Inclination (inc)	89.4 +/- 1.72
Airmass coefficient 1 (a1)	4.76 +/- 2.16
Airmass coefficient 2 (a2)	-1.39 +/- 1.53
Scatter in the residuals of the lightcurve fit is	39.07 %
Best Comparison Star	None
Optimal Aperture	1.7
Optimal Annulus	4.0
Transit Duration (day)	0.0488 +/- 0.0073

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.231 ± 0.068 vs 0.1714 (deviation 0.79σ)
Duration fit vs published	0.0488 d vs 0.0892 d (deviation 4.98σ)
Mid-time O–C	8.7 min
Depth SNR	3.1
Residual scatter	39.07 %

Light curve

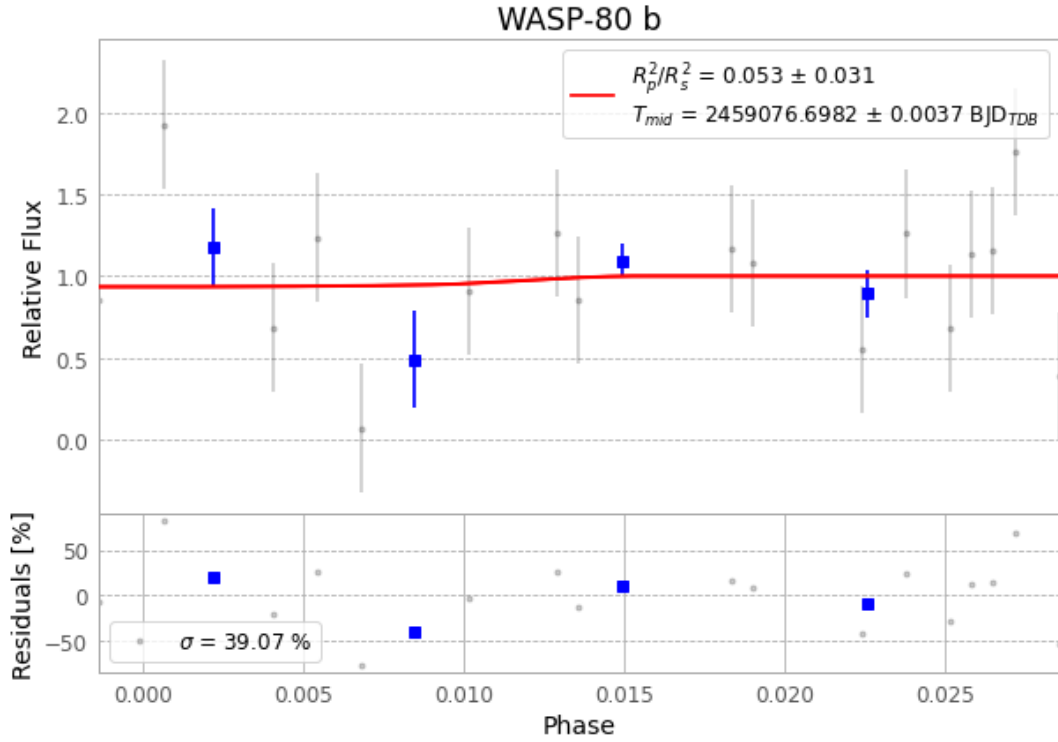


Figure 1 — FinalLightCurve_WASP-80 b_2020-08-14.png (SHA-256 55c11a5c86ec6a6c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [464, 158], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1b6f861b7a2d264a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

44 FITS frames (29 MB), WASP-80200815043014.FITS → WASP-80200815064214.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-80-200815.log (7e4902581f47...), FinalLightCurve_WASP-80 b_2020-08-14.png (55c11a5c86ec...), FinalLightCurve_WASP-80 b_2020-08-14.csv (51f7e7ab7937...)

Compute resources

Wall time 92.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 23:48Z.